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Abstract

Online learning is a growing area in occupational therapy education. It is important for faculty to use the modality of online learning to understand the influence of student personality traits, defined as relatively stable characteristics that influence thinking and behavior, on student satisfaction and success in graduate level programs. By understanding the influence of personality traits, faculty can better develop online materials that are flexible enough to support individuals who may have a higher risk of low satisfaction and success. There has been plentiful research in the areas of education and psychology assessing the influence of personality on the online learning experience. The goal of this systematic review was to collect, evaluate, and synthesize this literature to provide guidance to faculty working in or developing online courses or programs in occupational therapy. This systematic review was conducted according to the Cochrane Collaboration methodology and reporting is consistent with the PRISMA guidelines. The article search included January 2000 through June 2024. After duplicates were removed 848 articles remained, ultimately 99 were eligible for a full text review, and 23 were included in the three themes which were synthesized based on design and type of online education. The review highlights what is currently known related to the influence of personality type on online education. The information can be used to help sensitize educators to the influence different personality types can have when teaching online courses.

Keywords

Systematic review, higher education, on-line education, personality traits

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The Influence of Student Personality Traits on Satisfaction and Success in Online Education in Higher Education: A Systematic Review

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ABSTRACT

Online learning is a growing area in occupational therapy education. It is important for faculty to use the modality of online learning to understand the influence of student personality traits, defined as relatively stable characteristics that influence thinking and behavior, on student satisfaction and success in graduate level programs. By understanding the influence of personality traits, faculty can better develop online materials that are flexible enough to support individuals who may have a higher risk of low satisfaction and success. There has been plentiful research in the areas of education and psychology assessing the influence of personality on the online learning experience. The goal of this systematic review was to collect, evaluate, and synthesize this literature to provide guidance to faculty working in or developing online courses or programs in occupational therapy. This systematic review was conducted according to the Cochrane Collaboration methodology and reporting is consistent with the PRISMA guidelines. The article search included January 2000 through June 2024. After duplicates were removed 848 articles remained, ultimately 99 were eligible for a full text review, and 23 were included in the three themes which were synthesized based on design and type of online education. The review highlights what is currently known related to the influence of personality type on online education. The information can be used to help sensitize educators to the influence different personality types can have when teaching online courses.

Introduction

The rise of online learning in higher education has provided a flexible learning environment for a wide range of students. However, the effectiveness of online instruction varies significantly based on individual student characteristics, particularly personality traits (Ezra et al., 2021). Demographic factors such as age, gender, learning preferences, and psychological traits play critical roles in determining student satisfaction and success in these settings (Lazinski et al., 2023). Among these factors, personality traits, which are defined as habitual patterns of behavior, thought, and emotion, have been identified as critical predictors of student success (Baruth & Cohen, 2023; Bruso et al., 2020; Irani et al., 2004; Tlili et al., 2023). Studies show traits such as conscientiousness, openness, and agreeableness positively correlate with higher engagement and achievement, while neuroticism can be associated with lower satisfaction and academic success in online environments (Baruth & Cohen, 2023; Bahçekapih & Karaman, 2020; Yu, 2021).

Online education, where students learn partly or completely virtually from an instructor, is becoming more prevalent. Online education takes different forms. Hybrid education, which blends in-person and online learning, is increasingly being adopted by graduate schools (Banning et al., 2021). In 2021, approximately 1.8 million postbaccalaureate students were enrolled in at least one type of online education course and 1.3 million will be enrolled solely in online education courses (National Center for Education Statistics, 2023). Therefore, understanding how personality traits contribute to student success in hybrid education is becoming increasingly important in occupational therapy education. This is especially relevant as 58% of occupational therapy doctorate (OTD) programs, 54% of Master of Occupational Therapy (MOT) programs, and 47% of occupational therapy assistant (OTA) programs reported offering up to 24% of their curriculum through distance education (completely online or hybrid), highlighting the growing trend of online learning (Harvison, 2024). Currently there is no systematic review that consolidates the literature addressing the impact of personality traits on success and satisfaction in online learning formats in higher education. This review aims to address the gap by synthesizing existing evidence to provide students and educators with a clearer understanding of how personality traits influence online learning experiences.

Online Learning: Advantages and Challenges

Online learning has rapidly surged in popularity due to its flexibility and accessibility, particularly for non-traditional and geographically dispersed learners (Ezra et al., 2021). It has also proven to be a viable, evidence-based approach for healthcare education (Kaye & Kim, 2023). Literature highlights challenges of online learning such as reduced student engagement, feelings of isolation, and difficulty maintaining motivation, all of which can significantly impact student success (Baruth & Cohen, 2023). In addition, online learning attracts more non-traditional students, who balance additional responsibilities, which could overall affect their performance (Lazinski et al., 2023). In this context, personality traits may either exacerbate or mitigate these challenges. There is limited research examining the impact of online learning within occupational therapy education, however other allied health professions educational research found

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online learning students may perform slightly better than their traditional peers (Benaroya et al., 2021; Howard, 2019; Pucillo et al., 2020). Therefore, it is important to gain a better understanding of online learning strategies to promote academic success.

Personality Traits and Educational Outcomes

The Big Five personality model (McCrae & Costa, 1985) categorizes personality traits into five domains: openness, conscientiousness, extraversion, agreeableness, and neuroticism. The Meyers-Briggs (Myers, 1962) assessment categorizes personality traits as dichotomies extraversion/introversion, sensing/intuition, thinking/feeling and judging/perceiving. These personality traits refer to a relatively stable characteristic within a person that describes their pattern of thinking, feeling and behavior (American Psychological Association [APA], 2025). Researchers widely apply these personality models in studies examining educational outcomes in both face-to-face and online learning environments (Tlili et al., 2023). For example, students high in conscientiousness on the Big Five personality model tend to be more disciplined and motivated, resulting in better performance and higher satisfaction with online learning (Bruso et al., 2020).

Studies have also shown extraversion and agreeableness, traits associated with interpersonal interactions, can affect satisfaction levels in online learning, where social engagement is limited (Irani et al., 2004). Meanwhile, students with high levels of neuroticism may experience greater stress, which negatively impacts their success and satisfaction (Tlili et al., 2023). This systematic review aims to compile and synthesize literature regarding personality traits and student outcomes in online learning to gain a better understanding of the relationship between these factors.

The Need for a Systematic Review

Systematic reviews are essential for consolidating evidence, evaluating the quality of studies, and providing accessible insights for decision-makers (Higgins et al., 2019). This review followed the Cochrane Collaboration methodology and adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines (Page et al., 2021). The research question guiding the review was: “What is the evidence reporting the influence of student personality traits on their satisfaction and success in online learning in higher education?”

Methods

Study Design

The systematic review was conducted following the Cochrane Collaboration methodology (Higgins et al., 2019) and reported it in accordance with the PRISMA Guidelines for systematic reviews (Page et al., 2021).

Search Strategy

The search terms were developed collaboratively by the review authors and a medical librarian who is also an author of this article. The search strategy was developed, and the search conducted by a medical librarian (see Supplemental Table 1). Both Medical

Subject Headings (MeSH) and text words were utilized, adapting them as necessary for each database. The following databases were searched: PubMed, Medline, PsycINFO, CINAHL, Cochrane Reviews, Web of Science, ERIC.

Eligibility

Inclusion Criteria

Inclusion criteria included peer reviewed studies with January 2000 through June 2024 publication dates, English language, evidence level 1b, 2b, 3b, or 4, as well as qualitative studies. Participants were students in any higher education program taking part in online education in some capacity. The outcomes of primary interest were satisfaction based on satisfaction surveys and success based on grades and/or completion. All studies must have included a recognized personality assessment. Personality assessments included were variations of the Big Five Type Assessments or Myers-Briggs assessments.

Exclusion Criteria

Exclusion criteria included papers that did not meet inclusion criteria such as non-peer reviewed publications (dissertations, theses, presentations), outside the dates of included publications, non-English language, and looking at programs in higher education without some sort of online component included. Articles were also excluded if they did not use a recognized personality assessment based on either the Big Five Type or Myers-Briggs assessments. Additionally, studies were not included if the instructor was at a distance and students were together or if they looked only at the influence of learning styles, assignment preferences, or self-regulated learning. Finally, there needed to be results related to satisfaction and/or success.

Levels of Evidence

Each article evaluated in the review was assigned a level of evidence based on the Oxford Centre for Evidence-Based Medicine (OCEBM; 2009) framework (see Table 1). Systematic reviews were not included in the analysis.

Table 1

Levels of Evidence

Level	Type of Evidence
1A	Systematic review of homogeneous RCTs (similar population, intervention, etc.)- with or without meta-analysis
1B	Well-designed individual RCT (Not a pilot or feasibility study with a small sample size)
2A	Systematic review of cohort studies
2B	Individual prospective cohort study, low quality RCT (e.g. <80% follow-up or low number of participants, pilot and feasibility studies), ecological studies and two group nonrandomized studies
3A	Systematic review of case-control studies
3B	Individual retrospective Case-control study, one group nonrandomized pre-

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	posttest study, cohort studies
4	Case series (and low-quality cohort and case-control study), single group descriptive
5	Expert opinion without explicit critical appraisal

Procedure

Selection Process

The selection process included three steps. First, the medical librarian conducted the database search and removed any duplicates. Second, four authors independently reviewed the abstracts using Covidence, a systematic review software. After the independent review, the authors met to discuss any differences in their evaluations and reached consensus on which studies to include or exclude. Third, the authors collected PDFs of the selected articles to move forward to a full-text review. Three teams of two reviewers each conducted full-text reviews in Covidence. The full group met to discuss differences and consensus was reached. Finally, the full texts that remained were divided into two teams of three reviewers and data was extracted using an evidence table including study authors, methods, risk of bias evaluation, participants, setting of intervention, interventions/academic program, controls, assessments/measurement, and outcomes (see Supplemental Table 2).

Risk of Bias Assessment

The authors evaluated the risk of bias using the Mixed Methods Appraisal Tool (MMAT) version 18 (Hong et al., 2018). This tool allowed for the appraisal of all types of research including quantitative descriptive studies. Two authors reviewed all articles for risk of bias and then reviewed together until there was a 100% agreement.

Analysis

The authors performed a qualitative analysis of the data extracted from the studies. The findings were analyzed and summarized to answer the stated review question. Results were divided into thematic groups and were analyzed in terms of “strength of evidence.” The strength of evidence was determined using a formula based on the number of studies, level of evidence, quality/risk of bias, and significance of the findings for each theme or subtheme (United States Preventive Task Force, 2018). This can result in a strong, moderate, or low strength of evidence/strength of certainty (see Supplemental Table 3).

Results

Study Registration

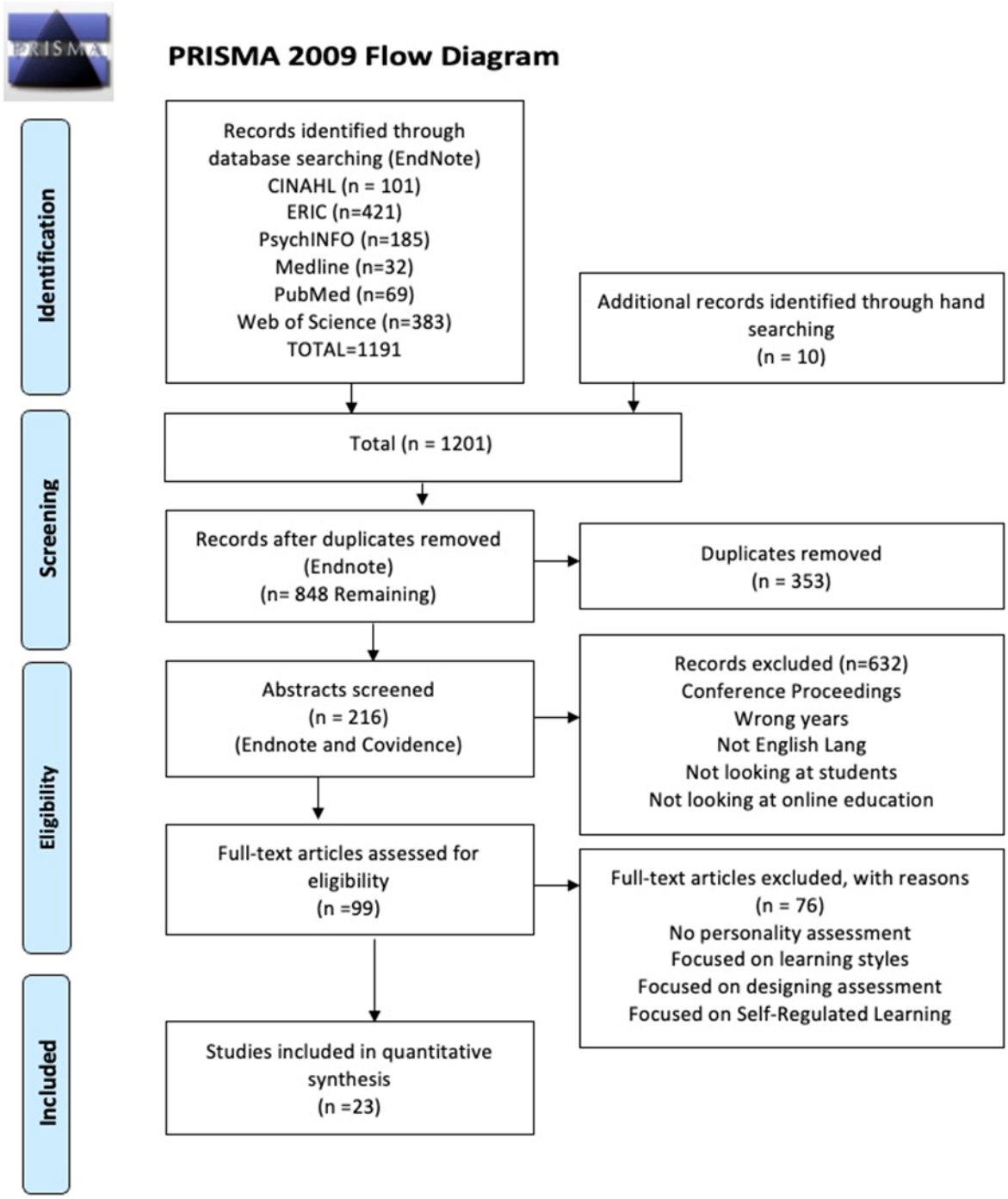
The protocol was registered with Open Science Framework (OSF).

Study Selection

The search resulted in 848 articles after duplicates were removed. After reviewing for appropriateness in terms of meeting inclusion criteria parameters, 23 remained in the final synthesis (see Supplemental Table 2). All the articles included were level 4 evidence. They are presented in the four themes synthesized based on design and type of online learning.

Figure 1

PRISMA Flow Diagram



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Study Characteristics

The included studies took place in a variety of countries and educational settings. The following tables show specific characteristics. Table 2 shows the higher education levels included in the studies.

Table 2

Student Grade Levels

Student Level	Number of Studies
Graduate	3
Undergraduate	9
Not specified	7
Mixed	4

Table 3 presents the various areas of study in the studies reviewed. There were no studies focused on occupational therapy.

Table 3

Course Content

Course content	Number of Studies
Agriculture Leadership	1
Business	1
eMBA	1
Engineering	1
Computer course in Education College	1
Psychology	4
Mixed	2
Technology	2
Not Specified	10

Table 4 illustrates the physical location of the students in the studies reviewed. There was a large international presence in the articles found and included.

Table 4

Location of Students

Location	Number of Studies
China	1
Columbia	1
Iran	2
Israel	2
Japan	1
Republic of North Macedonia	1
Taiwan	1
Thailand	1
Turkey	3
United State of America	10

Program Characteristics

Online education experiences varied from fully online academic programs, multiple online courses, a single online course, or a single online learning experience. Table 5 provides an overview of the different program characteristics in the studies. Specifically, it shows the level of online experience for the students.

Table 5

Program Characteristics

Program Type	Number of Studies	Studies
Fully Online Course	15	Alkis & Taskaya Temizel, 2018; Bahçekapili & Karaman, 2020; Baker & Moyer, 2019; Baruth & Cohen, 2023; Bhagat et al., 2019; Bishop-Clark et al., 2007; Bolliger & Erichsen, 2013; Chesser et al., 2020; Daughenbaugh et al., 2002; Harrington & Loffredo, 2010; Idrizi et al., 2021; Irani et al., 2004; Keller & Karau, 2013; Schniederjans & Kim, 2005; Wu & Yu, 2023
Fully Online Program or University	3	Bayram et al., 2008; Pinzón & Para Guarnizo, 2022; Tabe Bordbar, 2013
Another Online Experience	3	Blau et al., 2017; Mosalanejad et al., 2010; Yamamoto et al., 2023
Online Experience	2	Downs, 2019; Watjatrakul, 2016
Undefined		

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Types of Personality Assessments

Different personality inventories were used among the different studies. There were two overarching assessments used in the studies, either a Big Five type or Myers-Briggs type. Table 6 shows the variety of personality assessments that were used.

Table 6

Personality Assessments

Personality Assessments Used	Number of Studies
Big Five Type Assessments	
Big Five Inventory Type [Big Five Inventory] (John & Srivastava, 1999)	11
Wanderlic 2004 (Mount et al., 1999)	1
Mini-International Personality Item Pool (Donnellan et al., 2006)	1
NEO PI-R (Costa & McCrae, 1989)	1
NEO Five Factor Inventory (FFI; Costa & McCrae, 1989)	3
Myers-Briggs Type Assessments	
Myers-Briggs Type Indicator (Myers, 1962)	4
Kiersey Temperament Sorter (Kiersey & Bates, 1978)	1

Types of Outcomes

The primary outcomes of interest in this review were satisfaction and success. Satisfaction was typically assessed through a satisfaction survey and success was assessed through grades and completion. Some studies did include secondary outcomes such as learning preferences, self-efficacy, and stress.

Risk of Bias Within Studies

Risk of bias was assessed and a full evaluation of risk of bias is reported for each article in Supplemental Table 4. The assessment was completed by two reviewers who worked together to reach 100% agreement in the ratings.

Synthesis of Results by Theme

Strength of Evidence

The following are brief syntheses of the studies included in the review. Overall, there were three themes with theme 1 and 2 having sub-themes. All the articles in this review were level 4 literature. This means, based on the United States Preventive Task Force (2018) strength of findings evaluation, all the themes below are low strength of evidence. The findings are still of interest and useful to educators in terms of understanding what the current evidence shows.

Theme 1. Interaction between Personality and Online Education. Theme one included cross-sectional descriptive studies that explored the interaction between personality and online education. This theme was further divided into five sub-themes based on the outcome of interest.

Sub-Theme 1: Grade Point Average (GPA). Five level IV studies addressed the outcome of GPA based on personality type and taking fully online course or program for university students (N=1542). The studies used a Big Five type of personality trait assessment, and the courses covered various disciplines including psychology, business, engineering, and others that were not specified. Three studies found that GPA was positively correlated with openness, agreeableness, conscientiousness, and emotional stability for fully online courses or programs (Bahçekapili & Karaman, 2020; Pinzón & Guarnizo, 2022; Schniederjans & Kim, 2005). Bahçekapili and Karaman (2020) evaluated a fully online course and found self-efficacy positively mediated the neuroticism variable and external locus of control negatively mediated conscientiousness. Neuroticism was found to negatively correlate with GPA in these three studies as well; however, Idrizi and colleagues (2021) found students assessed with higher neuroticism performed better in online courses. Bayram and colleagues (2008) found introverts preferred online courses. Pinzón and Guarnizo (2022) connected demographics with their outcomes and found open-mindedness strongly influenced GPA, with open-minded men achieving the highest scores. They also found agreeableness scores were significantly lower for those under 24 than those over the age of 33. None of the five studies found extraversion influenced GPA.

Sub-Theme 2: GPA and Satisfaction. Two studies assessed both GPA and satisfaction and addressed personality traits using the Meyers-Briggs assessment. Most students were undergraduates taking part in fully online courses at universities in the United States (N=195). Courses included nursing, psychology, and education. Neither of these studies found a significant effect of personality on GPA but did influence satisfaction (Bishop-Clark et al., 2007; Harrington & Loffredo, 2010). Harrington and Loffredo (2010) found introverts preferred online classes and extroverts preferred face to face. Bishop-Clark and colleagues (2007) found personality did influence scores on specific types of assignments, as introverts and thinkers did better on quizzes than extraverts or feelers. Extraverts and intuitives scored higher than introverts and sensing types on discussion assignments. Finally sensing/thinking students were more likely to recommend the course and more satisfied than intuitive/feelers as intuitive/feelers felt more isolated in online courses.

Sub-Theme 3: Satisfaction. Four level 4 studies assessed satisfaction with fully online courses based on personality (Baker & Moyer, 2019; Bhagat et al., 2019; Keller & Karau, 2013) with one study not defining the extent of online education participants completed (Downs, 2019). All studies used a Big Five type of personality trait assessment. Participants were undergraduate and graduate students (N=862) taking part in any online class. In all studies, conscientiousness positively influenced satisfaction with online education. Students with high conscientiousness had less anxiety and frustration and were more likely to recommend others to take the class

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(Baker & Moyer, 2019; Bhagat et al., 2019; Downs, 2019; Keller & Karau, 2013). Baker and Moyer (2019) found intrinsically motivated students were more engaged, learned better, were more satisfied with online courses, and were less anxious and frustrated. Keller and Karau (2013) found agreeableness and openness significantly and positively related to engagement and satisfaction, while extraversion did not affect satisfaction with online courses. Agreeableness and conscientiousness were found to positively influence satisfaction with instructors' skills while neuroticism was negatively related (Bhagat et al., 2019). Finally, Downs (2019) found introverts preferred starting with in depth learning strategies and extraverts preferred starting with breadth and moving into depth.

Sub-Theme 4: Assignment Preferences. Two level 4 studies assessed how personality influenced assignment preferences in fully online courses (Baruth & Cohen, 2023; Daughenbaugh et al., 2002). The participants were graduate and undergraduate students (N=254). One study used a Big Five type of assessment (Baruth & Cohen, 2023) and one used a Meyers-Briggs type of assessment (Daughenbaugh et al., 2002). Baruth and Cohen (2023) found there was a significant correlation between extraversion and discussion groups and the use of different types of media. They also found agreeableness scores resulted in significantly higher satisfaction with all forms of assignments. For students high in conscientiousness, they observed a positive correlation in terms of satisfaction with media assignments, and for students high in openness, they found a high correlation of satisfaction with discussion groups. Daughenbaugh and colleagues (2002) found extroverts preferred interactive online elements (chat rooms, discussions) and introverts participated more via email.

Sub-Theme 5: Learning Engagement. One level 4 study addressed the interaction of personality type with learning engagement in a fully online course (Wu & Yu, 2023). This study included undergraduate and graduate students taking online courses (N=1004). They found extroversion, agreeableness, openness to new experiences, and conscientiousness had a positive association with online learning engagement. Neuroticism was the only personality type that had a negative relationship. Adaptability mediated the level of engagement for those who were assessed with extroversion, agreeableness, and openness to new experiences. Students with less adaptability did not show the same positive association with engagement. Adaptability did not influence the findings related to neuroticism or conscientiousness.

Theme 2. Comparing the Interaction of Personality on Multiple Different Class Formats. Theme 2 included studies that compared a variety of class formats and how student personality traits interact with them. This theme has three sub-themes based on the outcomes of interest.

Sub-Theme 1: Satisfaction. Two level 4 studies addressed the outcome of satisfaction while comparing the choice of type of class format the students selected. Using a Big Five assessment, one study focused solely on the influence of extraversion (Yamamoto et al., 2023). The participants were undergraduate university students in Japan (N=175) who had taken part in one of the following class formats: task based,

asynchronous online, synchronous online, and face to face. The study found extraversion resulted in higher satisfaction among those who took part in face-to-face classes.

One study used the Meyers Briggs assessment and looked at satisfaction among graduate students (N=72) in the United States enrolled in multiple courses either in online or blended formats (Bollinger & Erichsen, 2013). Participants were generally satisfied with both teaching formats. For those in the fully online course, 81% stated they could not have taken the course if it was not online. Extraverts were less likely than introverts to recommend the fully online course, and sensors were more satisfied working independently than intuitive learners. Feelers and intuitives in the online course were more satisfied with threaded discussions and interactions. Finally, introverts and intuitives were more dissatisfied with the blended course (Bollinger & Erichsen, 2013).

Sub-Theme 2: Grades. Two level 4 studies assessed the outcome of grades only. Both studies involved undergraduate students (N=357) and used a Big Five personality type assessment. One study compared asynchronous to blended settings for an introductory IT class (Alkis & Taskaya Temizel, 2018), and the other compared blended and face to face formats in a psychology class (Mosalanejad et al., 2010). In both studies, high conscientiousness was significantly related to high grades, no matter the course delivery format. Alkis and Taskaya Temizel (2018) also found self-efficacy significantly mediated course grades in the online course setting but not the blended class. Teaching format, students' final scores, and personality type had a significant relationship between personality traits of openness, extraversion, agreeableness, and conscientiousness, but not neuroticism (Mosalanejad et al., 2010).

Sub-Theme 3: Grades and Satisfaction. One level 4 study assessed grades and satisfaction for one learning experience in multiple settings (Blau et al., 2017). Blau and colleagues looked at two spectrums, extrovert/introvert and neurotic/emotionally stable undergraduates (N=76), who were randomly assigned to one of three settings: synchronous two-way session, synchronous one-way session, and face to face. Each group took part in one 25-minute lesson taught by the same teacher using the different delivery methods. Significant findings with a high effect size showed neurotic students enjoyed and performed better in a face-to-face setting while emotionally stable students were successful in all of the learning formats and had significantly higher scores. Extroverts preferred the face-to-face learning environment but had lower grades in that condition.

Theme 3: Evaluating the Interaction of Personality Trait and Student Choice of Class Format. The third theme which emerged was evaluating the interaction of personality and choice of class type, or why students may have chosen to be online versus in-person for an entire course (Chesser et al., 2020), program (Tabe Bordbar, 2013), or had an undefined online experience (Watjatrakul, 2016). There were three level 4 studies in this theme with all looking at undergraduate students (N=673). Outcome assessments included the Big Five Inventory, NEO-Five Factor Inventory, grades, and students' choice or perceptions around enrolling in online versus in-person

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courses. Geographically, the studies were completed in the United States (Chesser et al., 2020), Thailand (Watjatrakul, 2016), and Iran (Tabe Bordbar, 2013). Students assessed with high level of conscientious, agreeableness, and openness to experience were more likely to adopt online education, with those assessed as neurotic being least likely. Students choosing an online course had a higher GPA than those enrolled in the traditional classroom, and conscientiousness was an indicator for both settings of a higher grade (Chesser et al., 2020).

Discussion

This review identified studies that explored how personality traits influenced success and satisfaction with online learning in higher education regardless of the specific field of study. The review highlighted two interesting facts. First, the studies were all low level of evidence (IV) which seems to be common in educational research but does not lend itself to providing information that can guide educational practice well. The second was the fact that none of the studies found examined occupational therapy programs. The lack of research related to online education in occupational therapy is an important gap. As discussed earlier, the continued growth of online education in occupational therapy should guide researchers to assess the influence of this educational modality on occupational therapy education and student outcomes in general. Personality traits are only one aspect but can be important.

The review highlighted and reinforced the findings of previous studies related to this topic. The Big Five model (McCrae & Costa, 1985) is the most common and recognizable model. Based on the Big Five, the neuroticism personality trait was identified as the most negatively related to success and satisfaction with online learning (Baruth & Cohen, 2023). In contrast, those possessing the personality trait of agreeableness or conscientiousness tended to be successful no matter the learning methods (Tlili et al., 2019). Interestingly, one study found those with the personality trait of extraversion preferred in person classes but performed better in online classes (Blau et al., 2017). Extroverts also participated more actively in online group discussions than introverts (Lee & Lee, 2006). Although individuals with neurotic traits may feel reluctant to post messages, they still engaged in discussions, possibly benefiting from the time online learning gave them to manage anxiety (Caspi et al., 2006).

As the review showed, educators must recognize that some personality traits help individuals overcome challenges in online education, while the absence of certain traits may hinder those who are otherwise expected to succeed. Traits such as self-efficacy and internal locus of control play critical roles in all educational settings but are particularly important for those in an online learning environment (Edisherashvili et al., 2022). Since personality traits and locus of control are thought to generally remain stable throughout life, educators can enhance success by tailoring flexible courses and assignments to accommodate for these factors (Soto, 2021). However, as revealed in the review, self-efficacy can be enhanced by supporting behaviors like self-regulated learning which can be developed and improved over time with support and practice (Bahçekapili & Karaman, 2020; Sitzmann & Ely, 2011). Additionally, the use of universal design in learning is a strong foundation providing flexibility for students regardless of

their personality traits. The concept of universal design transforms education by providing principles and strategies to design learning environments that suit all learners (Meyer et al., 2014).

Outcomes of interest were assessed through GPA (success) and satisfaction surveys. The satisfaction surveys varied greatly, with the majority being developed by the researchers. They were typically labeled satisfaction surveys, but some went into more depth by including student perceptions toward online learning and student checklists/inventories related to satisfaction. Some only included a single question related to satisfaction. Outcomes of interest that emerged but were not part of our review question included anxiety, self-efficacy, locus of control, motivational strategies, perceived/subjective learning, and education attitudes. The lack of standardization of satisfaction surveys can be seen as another weakness in the study outcomes.
The review highlighted how personality factors affected satisfaction and academic performance, such as assignment scores or GPA. Other factors, like age, emerged to a lesser degree. Older students tended to perform better in online education than younger students (Baruth et al., 2021). This finding may reflect student maturity improving self-regulation skills and abilities to engage as an adult learner. Regardless, there are a myriad of student factors that influence success, and educators need to consider these factors during program and course development. The findings support the need for development of evidence-based, personality-informed, educational experiences tailored to enhance student satisfaction and success.

Limitations

The main limitation in this review was the level of evidence for the studies included. Having only level IV studies did not allow for anything but describing what the different programs showed. Future research should move beyond descriptive studies and begin to focus on implementing best practice interventions to assess their potential benefits for developing online courses that can be tailored to all students. Another limitation was the search only included studies from January 2000 to June 2024. While there is research related to this topic before 2000 the importance of online education has been growing rapidly in the 2000s. Additionally, the rapid evolution of technology since 2000 provides vastly different experiences of online learners in the earliest studies included in this review compared to the later years. While a university librarian was used to help with the search and Covidence to help with the evaluation, there is always the possibility of omitting articles that would have fit the inclusion criteria. Additionally, the studies came from various geographic locations, which may limit the applicability of their findings across different cultures. Also, differences in language may have led to some terminology not being universally understood.

Future Directions

Future studies need to develop and test interventions to increase the depth of knowledge and to move the field forward. This could include incorporating specific aspects of courses which support students with neurotic and extroverted traits in engaging more in online education. For example, evaluating the use of a mock schedule for learning to support students with neurotic traits or a collaborative learning

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activity to engage extroverted students. Another systematic review looking at what interventions have been used to address self-regulated learning and online learning success would be a natural next step and could guide the development of intervention studies. As mentioned, no studies related to occupational therapy education specifically were found. This is an area for future research as more and more occupational therapy education is being delivered in a hybrid fashion.

Implications for Occupational Therapy Education

Based on most of the studies examined in this review, it is more likely that students who are open, agreeable, conscientious, and emotionally stable will perform better in an online environment. Therefore, building in activities which support extroverted or neurotic students in being successful, such as a synchronous discussion or organizational supports, may allow these learners to be academically successful, engaged, and satisfied with the assignments and program. Understanding that locus of control and self-efficacy can mediate some of the challenges in an online setting, means faculty can identify students who may need support in these areas to be successful.

Self-regulation plays a vital role in academic success and extends beyond education to both mental and behavioral health at all ages. Educators are uniquely positioned to foster self-regulation, which can improve outcomes in online health professional education. Studies have shown that students who possess strong self-regulation skills are more likely to succeed in online learning environments (Pucillo & Perez, 2023; Tunningley, 2021). Findings from previous studies have shown a significant predictive relationship for students with stronger self-regulation, exhibiting greater success with online health professional education (Pucillo & Perez, 2023; Tunningley, 2021). Self-regulated learning support is crucial for the online learning context. Educators must help students develop these skills.

Additionally, common educational practices, such as Universal Design for Learning (UDL), benefit all students, regardless of their personality traits, and contribute to their success in online learning (CAST, 2024). Educators must design courses that address diverse student learning needs. Universal design principles ensure inclusivity by creating flexible and adaptive learning environments that cater to various personality types and learning preferences (CAST, 2024).

Conclusion

The number of online occupational therapy programs is increasing, and the field is well-positioned to develop programs that are accessible to the broadest number of students possible. While personality type may not always be an educator’s first consideration, it is an important factor that educators should incorporate when developing the most supportive programs for all students. As traits are often seen as fixed the review guides educators to explore methods to tailor education to fit the needs of the student through UDL and to build self-regulated learning skills. This review provides information that can guide educational practice as well as guide future research in the many areas where there are currently gaps.

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Supplemental Materials

Supplemental Table 1

Example Search Strategy

MedLINE Search String
((“higher education”) OR (MH “Universities") OR (“graduate student”) OR (“undergraduate student”) OR (“college student”) OR (“college”) OR (“university”) OR (“Graduate education”) OR (“Undergraduate education”) OR (“College education”) OR (“University education”)) AND ((MH "personality") OR (MH “Personality Tests”) OR (“personality”) OR (“personality traits”) OR (“Personality measures”) OR (“5-factor model”) OR (“big 5 inventory) OR (MH "personality development") OR (“myers-briggs”) OR ("Keirsey Temperament Sorter") OR (“Extravert*”) OR (“Extrovert*”) OR (“agreeableness”) OR (“openness”) OR (“conscientious*”) OR (“neurotic*”) OR (“introvert”) OR (“extrovert”) OR (MH "resilience, psychological") OR (“learning styles”) OR (“TPLS”) OR (“self-regulated learning”) OR (“Self-regulated”) OR (“Self-regulated learner”) OR (“effective educational experiences”) OR (“belonging”) OR (“independent learners”)) AND ((MH“Education, Distance”) OR (“online education”) OR (“hybrid education”) OR (“remote education”) OR (“distance education”) OR (“elearning”) OR (“online course”) OR (“virtual course”) (OR) (“online instruction”) OR (“virtual learning”) OR (“Virtual instruction”) OR (“distance learning”) OR (“virtual education”) OR (“e-learning")) AND ((“Academic Performance”) (“student responsibility”) OR (“Student attitudes”) OR (“student satisfaction”) OR ((“Stress”) OR (“stress perception”) OR (“satisfaction”) OR (“success”) OR (“social presence”) OR (“student engagement”) OR (MH “Academic Success”))

Supplemental Table 2

Evidence Extraction Table

Theme 1: Cross sectional descriptive studies exploring interaction of personality and online education				
Title/Author/ Year/Country/ Level of Evidence/Study Design/ Risk of Bias (quality assessment)	Participants (# participants, mean age, gender percentages)/ Inclusion Criteria/study setting	Intervention and Control Or Study Description	Outcome Measures of Interest	Results
Sub-Theme 1: Big Five Type Personality Assessments				
Personality and Satisfaction with Online Courses: The Relation between the Big Five Personality Traits and Satisfaction with Online Learning Activities Baruth & Cohen, 2023 Israel Level 4 cross sectional	Participants: N=108 Age: NR; Gender: NR Inclusion criteria: University students enrolled in an accredited online academic course. Study setting: Online course in a large, accredited university in Israel	Study Description: Students participated in an online learning course. The tested techno-pedagogical learning solutions (TPLS; discussion groups, digital books, online assignments, surveys/polls, media) were designed especially for this online course and offered to the students during all the course lessons. Each activity was designed for specific learning content.	-Big-Five Inventory (BFI) (John & Srivastava, 1999) -Questionnaires satisfaction with TPLS used	Results: - Significant and high correlation found between extraversion and satisfaction with discussion groups -Moderate correlations found between extraversion and media and assignments - Significant high correlations between agreeableness and all satisfaction measures: general satisfaction discussion groups and media - Strong positive correlation with media was also found for conscientiousness-Strong positive correlation with openness and discussion group satisfaction -Weak moderate correlation with general openness and satisfaction), surveys and polls and assignments
Predictor variables of	Participants: N= 354 age range 18-	Study Description: - to identify the	-Demographic -BIG FIVE	Results -GPA is explained by sex, age, the

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academic achievement for psychology students in distance education Pinzón & Guarnizo, 2022 Colombia Level 4 cross sectional	52, M = 29; female 86.72% Inclusion criteria: Students studying psychology remotely at a private distance education university in Bogota, Columbia. Study setting: university in Bogota, Columbia	predictive role of sociodemographic variables, learning styles, motivation and personality traits on GPA. -distance psychology students who were recruited from a stratified probability sampling, voluntarily participated in the online survey.	Questionnaire (BFQ, Caprara et al., 2007). -GPA	personality traits tenacity, and open mindedness -Demographics: Significantly higher scores for males on open-mindedness. - Significantly lower agreeableness scores for ages <24, compared to >33
Relationship between university students' personalities and e-learning engagement mediated by achievement, emotions, and adaptability Wu & Yu, 2023 China Level 4 cross sectional	Participants: N=1004 age range 17-28; Females 77.3%; 65% undergraduates, Inclusion criteria: Taking part in online education. Enrolled in Guizhou University in China. Study setting: Guizhou University in China	Study Description: -Online survey and in person semi structured interviews with students who have experience in online education. -The present study used the five-factor model of personality as the main theoretical framework to explore how students with different personality traits engaged in e-learning.	- Personality- Big Five Inventory -Scale to measure e learning engagement -Adaptability scale	Results: -extroversion, agreeableness, openness to new experiences, and conscientiousness exhibited positive associations with e-learning engagement -neuroticism demonstrated a negative relationship with e-learning engagement -adaptability played a mediating role in the relationship between personality traits (excluding conscientiousness and neuroticism) and e-learning engagement
A Path Analysis of Five-Factor Personality Traits,	Participants: N=525 ages 19-59 years, M = 30.9 years;	Study Description: The participants in both universities attend a	-Five-factor model of personality scale	Results: -personality is a predictor of academic performance in both

Self-Efficacy, Academic Locus of Control and Academic Achievement among Online Students Bahçekapili & Karaman, 2020 Turkey Level 4 cross sectional	38% female Inclusion criteria: Voluntary participation of students from the two selected universities Study setting: Two different universities in Turkey online courses	distance education program. The participants attend their classes in an online live class environment at a planned time. The course utilizes an online whiteboard, presentation, other materials in online learning environments, vocally communicate during the course, and use chat features to engage as well.	-Academic locus of control scale -General self-efficacy scale -Grade Point Average (GPA)	online and blended course settings. -A significant positive relation between the conscientiousness trait and course grades in both settings. - general self-efficacy and external academic locus of control variables are mediating the conscientiousness and neuroticism variables; -only general self-efficacy is mediating the openness personality trait.
The Relationship between Students' Characteristics and Their Impressions of Online Courses Baker & Moyer, 2019 United States Level: 4 cross sectional	Participants: (N=84; Ages: 18-59; M = 25.5; Gender: 100% female; 61% unmarried, 73% commuted to campus, 69% were juniors or seniors, 11% were graduate students, 44% were full-time Inclusion criteria: undergraduate or graduate university students taking a mix of in-class and online courses (1 online course required for participation)	Study Description: The study included students taking a mix of in person and online courses (1 online course required for participation) -Fifty-two percent of the participants reported that online courses were typically required for their major or minor course of study. -Most participants took either seven or eight weeks (35%) or 14-week (44%) online courses, but other lengths of online courses (i.e., 6 and 10	-Big Five Inventory (BFI) for measuring conscientiousness -Online Course Impressions (OCI) instrument	Results: -Age: participants 25 years or older had sig. higher levels of conscientiousness - Association between conscientiousness, motivation and social presence on students' impressions and satisfaction with online learning. -Conscientious students more engaged, perceived value, recommended others to take online courses, and preferred online courses. - Students with high conscientiousness less anxious and frustrated. - Intrinsically motivated students more engaged, perceived value, recommended others to take online

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	Study setting: students at a small, private women's liberal arts college in the United States.	weeks)		courses, learned better and preferred online courses, and became less anxious and frustrated.
The importance of personality in students' perceptions of the online learning experience Keller & Karau, 2013 United States Level: 4 cross sectional	Participants: N=250 online students; mean age= 35, age range 19- 57; 73% female; 65.2% employed full time; 65.2% were married; 59% undergraduates. Inclusion criteria: students enrolled in one or more online courses at a traditional, state-funded university in the southeast of the United States Study setting: online courses at a state-funded university in the United States	Study Description: -Examined the relationship between the Big Five personality dimensions and five specific types of online course impressions (engagement, value to career, overall evaluation, anxiety/frustration, and preference for online courses). -Conducted a survey.	-Big Five- 50-question version of the lexical Big-Five Factor structure (Goldberg,1992) -Online course impressions (OCI)	RESULTS: -Conscientiousness positively related to engagement, value to career, overall evaluation, and preference for online courses, and was negatively related to anxiety/frustration. -Agreeableness and openness to experience were both positively and significantly related to engagement, value to career, and overall evaluation. -Significant positive relationship was also found with emotional stability and value to career. - No significant relationships were found between extraversion and any of the specific values. -Students with higher levels of Conscientiousness, work experience, and marital status were significantly less anxious or frustrated with online courses. -Conscientiousness and academic level (undergraduate or graduate) were significant individual predictors. With undergraduates expressing strong support for online courses.
The Role of	Participants: N=108	Study Description:	-Big Five	Results:

Personality Traits in Web Based Education Bayram et al., 2008 Turkey Level: 4 cross sectional	Age: NR; Gender: NR Inclusion criteria: learners randomly chosen out of 570 students enrolled in the eMBA master's degree of Bilgi University in 2005 Study setting: Turkish University Web based master's degree	-The purpose of the study was to find the relationship between students' personality traits and their satisfaction level with the courses Techno-pedagogical learning solutions (TPLS) as well as the course in general. -Questionnaires to measure personality types and satisfaction were administered to the e-MBA degree students when they came to campus for final exams	Inventory (1999) -Web Based Education Attitudes Scale -GPA	-Significant positive correlations were found between personality traits and attitudes towards web-based education. - Introverts prefer online courses -Extraverts prefer face-to-face -Non-personality reasons were biggest drivers of reasoning for selecting a specific type of education (schedule, travel, enjoyment in computer technology, and interest in innovation)
Relationship of Student Undergraduate Achievement and Personality Characteristics in a Total Web-Based Environment: An Empirical Study Schniederjans & Kim, 2005 United States Level 4	Participants: N = 260 Age Range:19-31 M= 22.1; Gender: NR Inclusion criteria: Enrolled in one of the two sections of undergraduate management information systems class for business majors Study setting: University of	Study Description: -This research study examines the relationship between student personality characteristics and their achievement scores as a means of identifying predictors of academic success. -The total web-based environment (TWE) course used in the study was self-paced and delivered via web (blackboard). Nothing was changed between	-Wonderlic® Personality Characteristics Inventory TM (PCI) survey assessment instrument (Barrick & Mount, 1991; Mount et al., 1999). -Course grades	Results: - Four basic personality characteristics (conscientiousness, openness to experience, emotional stability, and agreeableness) are highly correlated to student achievement in Web-based courses. -Extraversion was not significantly correlated to student achievement scores.

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cross sectional	Nebraska-Lincoln	the semesters when data was collected. -a group of 260 final participants were randomly taken from the 500 students.		
The impact of personality on students' perceptions towards online learning Bhagat et al., 2019 Taiwan Level 4 cross sectional	Participants: N=208 Age range 18-45; 54% female n= 112 Inclusion criteria: Students (undergraduate and graduate) from Taiwan with online course experience Study setting: University in Taiwan.	Study Description: - The aim of this exploratory study was to examine the impact of five personality traits (extraversion, agreeableness, conscientiousness, neuroticism, and intellect/imagination (openness to experience)) on the perception of students towards online learning. -Online survey of students, from undergraduate through PhD studies, attending a university in Taiwan, who had previously engaged in an online course.	-Mini-International Personality Item Pool (Mini-IPIP)- to access items of the big five model - Perception of Students towards Online Learning	Results: - Two personality traits (conscientiousness and openness) had a larger positive impact on students' perceptions towards online learning -neuroticism had significantly negative effects on participants of online courses. -Agreeableness, conscientiousness, and neuroticism significantly influenced perceptions of instructor skills. Agreeableness was the most significant personality type to influence this. -Extraversion, agreeableness, and conscientiousness significantly influenced perceptions of social presence/connectedness. Extraversion was the most significant predictor. -Agreeableness, and conscientiousness, significantly influenced perceptions of instructional design. Conscientiousness was the most significant influence. -Neuroticism significantly negatively influenced feeling of trust.

Analysis of Success Indicators in Online Learning Idrizi et al., 2021 Republic of North Macedonia Level 4 Cross sectional	Participants: N=155 (n-101 did not finish); N=54 students completed the case study and took the final test Age: NR; 39% female Inclusion criteria: University engineering students who were part of two separate online courses that had different levels of difficulty. Study setting: Saints Cyril and Methodius University, Skopje, Republic of North Macedonia	Study Description: Students were asked to attend two online courses, with different difficulty levels, during one semester. Analysis was conducted to determine which factors are significant for the academic performance of students taking online courses, as well as for their overall academic success. Intervention 1: Group A students (n=27); - C1 course were asked to pick their preferred type of learning materials; - C2 students were assigned the delivered materials by the instructor. Intervention 2: Group B students (n=27) -C1 course instructor assigned the type of delivered materials to group B. -C2 course, group B students picked their learning materials,	-NEO personality inventory -Student scores on tests within online course and cumulative GPA	RESULTS: -In online courses, Course difficulty and Group Affiliation were noted to have statistically significant impacts on overall test scores, but not overall academic success -Overall academic success, for all students: gender, type of provided materials, type of preferred materials, neuroticism, and extraversion were significantly connected to GPA. -Student assessed with Neuroticism did somewhat better in online classes. This suggests that students who rank higher on Neuroticism struggle in all educational settings but marginally less in online classes where they are able to manage their anxieties
Title: An	Participants: N=320	Study description: To	-NEO Five-Factor	Results:

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Exploration of the Relationship Between Personality Type and Satisfaction with Online Learning Environments Downs, 2019 United States Level 4 cross sectional	Age: NR; Gender: NR Inclusion criteria: Undergraduate students enrolled in online courses at the university. Study setting: A large public university in the United States	investigate the relationship between personality types and student satisfaction in online learning environments, and to determine how different personality traits influence students' satisfaction with online learning. -Students completed surveys to measure personality across five dimensions and student satisfaction related to online learning.	Inventory-3 -Student satisfaction survey	- Conscientiousness had higher satisfaction and neuroticism had lower satisfaction with online course. -Introverted students prefer starting with in depth-first learning strategies and extraverted students prefer starting with breadth-first, before going in depth in the topic
Sub-Theme 2: Myers-Briggs Type Indicator Assessment				
Title: MBTI personality type and other factors that relate to preference for online versus face-to-face instruction Harrington & Loffredo, 2010 United States Level 4 cross sectional	Participants: N= 166. 58% 26-45; 91% female; 72.3% undergraduates; 58% Caucasian. Inclusion criteria: University students enrolled in psychology, nursing, and education online courses at a large university in the United States, Undergraduate Study setting: large public university in	Study Description: - Assesses preferred online to face-to-face courses including personality to enhance online instruction for students whose primary preference is face-to-face instruction -Students completed an online consent for, a demographic survey, a questionnaire asking their general preference for online instruction vs face-to-face instruction, and a set of follow-up questions designed to	-Personality Assessment: Myers-Briggs Type Indicator (MBTI) -Academic Performance: Measured using GPA -self-reported satisfaction with online learning	Results: - 50% of the students preferred online and 50% preferred face to face. -Majority of introverts preferred online classes -Extraverts preferred face to face - Non-significant trend of perceiving types preferring face-to-face - Those who preferred online was due to convenience, enjoyment of technology, a desire for innovation -Those who preferred face-to-face classes was due to need to learn through listening, desire to better gauge emotional reactions of others.

	the United States	assess the reasons why they prefer one delivery mode over the other. Finally, they completed an online version of the MBTI Form M.		
The Effects of Personality Type on Web-Based Distance Learning Bishop-Clark et al., 2007 United States Level: 4 cross sectional	Participants: N=29 Age: NR; Gender: NR Inclusion criteria: Undergraduate students enrolled in an Introduction to Psychology course at a regional campus of a medium-size Midwestern University Study setting: Personality style inventory was completed in-person during initial face-to-face engagement	Study Description: Investigated whether student performance and satisfaction in an asynchronous, Web-based distance learning course differed based on personality factors (as measured by the Myers Briggs Type Indicator). -Initial face-to-face class followed by online course engagement (6 weeks in summer; 10 weeks in fall) taught online by the same instructor utilizing 13 modules, also met face-to-face for four separate class exams	- Myers Briggs Type Indicator: Personality -Grades - Flashlight's Current Student Inventory: student satisfaction	Results: -Personality had little relationship with performance but did influence satisfaction. -Introverts performed better than extraverts on quizzes -Thinkers scored better on quizzes than feelers. -The discussion score, extraverts scored higher than introverts and intuiting types scored higher than sensing types. -Sensing/thinking students were more likely to recommend the course to others and more satisfied than intuitive/ feelers - Intuitive/feelers were much more likely to feel isolated from other students than the sensing/thinking students reported
Personality Type and It's Relationship to Distance Education, Student's Course Perceptions and Performance	Participants: N=39; M age= 33 years, age range 21-54; 66% female Inclusion criteria: university students in a distance education course (agricultural	Study Description: Relationships between students' course perceptions and performance, and personality when taking part in distance education using two-way videoconferencing	-Myers-Briggs Type Indicator (MBTI) -perceptions of the course -demographics	RESULTS: -Instructional technique perceptions were strongly correlated with course grade and GPA for most personality types (extravert, introvert, intuitive, sensing, judging). -No significant correlation for thinking and perceiving types.

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Irani et al., 2004 United States Level: 4 Retrospective Case Study	leadership). Two-way videoconferencing Study setting: University in the United States	(Synchronous). -secondary data collected was course evaluations, and results from a Myers-Briggs personality type inventory, administered to students in the class during the term in which the course ran, were collected		-Gender influenced perceptions of instructional delivery and course management. -Age was correlated with course grade, GPA, and perceptions of instructional technique, course management, and social interaction. -Each personality type showed unique correlations, with introverts and extraverts having the strongest associations.
Personality Type and Online versus In-Class Course Satisfaction Daughenbaugh et al., 2002 United States Level 4 cross sectional	Participants: N=146, Age: NR; 78% female; 53% graduate students; 18.5% enrolled online, 81.5% enrolled in face-to-face course Inclusion criteria: Undergraduate or graduate student enrolled in online and face-to-face introductory computer courses in the College of Education Study setting: College of Education at the University of South Alabama.	Study Description: To determine if different personality types express more or less satisfaction with courses delivered online vs in the classroom. -Students were given 2 questionnaires that were web-based to rate subjects on four variables (extroversion or introversion, intuition or sensing, thinking or feeling, judging or perception) and course satisfaction.	-Kiersey Temperament Sorter (KTS) Instrument (similar to MBTI) -Course satisfaction instrument (CSI) that researchers developed	Results: -Extroverts preferred interactive elements (chat rooms, discussions) -Introverts participated more via email. -Sensing and intuitive types differed in course satisfaction. -Intuitive types favored online courses over traditional classrooms. -Perception group valued student interaction more than the judging group. -No significant satisfaction difference between thinking and feeling types. -In-class students reported higher satisfaction with interaction and environment than online students. -Males and females had nearly identical preferences for online vs. in-class courses.
Theme 2: Comparing the interaction of personality on multiple different class formats				

Sub-Theme 1: Big Five Type Personality Assessments				
Relationship of personality traits with degree of subjective learning and satisfaction in different class formats among Japanese university students Yamamoto, et al., 2023 Japan Level 4 Cross sectional	Participants: Start of project N=175 Final data used in analysis: N=150 participants 43% women; M age = 19.21 Inclusion criteria: university students Study setting: Japanese university, recruited through a web-based educational support system targeted at class attendees.	Study Description: -Examined the relationship of personality traits with degree of subjective learning and satisfaction in different class formats -The students completed an online questionnaire. -Class formats offered were: Task based On demand (online) Real time interactive (online) Face to face	- Short Form of the Big Five Scale-extraversion subscale only (Nami-kawa et al., 2012). -Reported if they participated in each type of class, then asked to evaluate each type of class they had participated in. - Subjective learning in each class format was measured using a 16-item scale - Satisfaction with each class format was a single question - Operation Anxiety subscale of the Computer Anxiety Scale (Hirata, 1990).	RESULTS: - different personality traits may influence undergraduate students' degree of subjective learning and satisfaction depending on the class format. - extraversion was associated with positive evaluations of face-to-face classes. -Computer anxiety was negatively related to evaluations of on-demand classes.
The Impact of Motivation and Personality on Academic Performance in Online and Blended Learning Environments	Participants: N=316 Average age: 22; 41% male. Inclusion criteria: undergraduate students who enrolled in an introductory IT course Study setting:	Project Description: Study Aims: Identify and compare students' motivation and personality traits on their academic performance in online and blended learning environments.	-Personability Traits: Big Five Inventory (BFI; (John & Srivastava, 1999) -Motivated Strategies for Learning Questionnaire (MSLQ) -Grades (final	Results: -A significant positive relation was found between the conscientiousness trait and course grades in both settings. - Conscientiousness trait was found to be positively and significantly related to frequency of LMS use in the online course setting. -Self-efficacy was found significantly

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Alkis, N., & Taskaya Temizel, 2018 Turkey Level: 4	Online and blended classes at Middle East Technical University (METU), Ankara, Turkey	Intervention 1: (n=189) (n=109 female, n=80 male; average age: 22.27) Online class (Asynchronous): Use online LMS to access the lecture notes, upload assignments and follow the announcements. 12-week session Intervention 2: (n=127) (n=76 female, n=51 male; average age: 22.03) Blended class: Students attend face-to-face lectures conducted in Computer laboratories. Each week, students attend two-hour laboratory sessions carried out by the instructors. 12-week session	course grade) -LMS use logs	related to both LMS use and course grades in the online course setting. No significant relation was found in the blended course setting. -A significant positive relation was found between task value and LMS use in the blended course setting.
How Do Medium Naturalness and Personality Traits Shape Academic Achievement	Participants: N=76 Age Range: 18-59, M=31.45; 58% female; 63% undergraduate Inclusion criteria: Students at the	-Participants were randomly assigned to 1 of 3 setting, in gender-balanced classroom-like groups of approximately seven participants. Intervention 1: (n=26) Two-way video	- Personality: Short Hebrew version of the NEO-PI-R Personality (tested for extroversion/ introversion and emotional stability/neuroticism	Results: -Neurotic students tended to enjoy and succeed more in face-to-face learning, emotionally stable students enjoyed and succeeded in all the learning conditions. -Extroverts tended to enjoy more natural learning environments but

and Perceived Learning? An Experimental Study of Face-to-Face and Synchronous E-Learning Blau et al, 2017 Israel Level: 4	Open University of Israel who were native Hebrew speakers, who volunteered to participate in the study. Study setting: Open University of Israel	conferencing lesson Intervention 2: (n=26) One-way videoconferencing lesson Intervention 3: (n=24) Traditional face-to-face lesson -Procedure: completion of the socio-demographic and personality traits questionnaires; participation in a lesson that was conducted according to one of the three experimental conditions described above. The lesson (~25 min), entitled 'Charisma of Politicians = Vision + Presentation' was delivered by an expert instructor, experienced in both face-to-face and synchronous teaching and unaware of the research hypotheses.	-Personality questionnaire (Blau and Barak 2012) - Perceived learning: self-report questionnaire - Knowledge acquired during a lecture: Achievement Test (11 question multiple-choice test) - Socio-demographic questionnaire	had lower achievements in these conditions. - Emotional stability versus neuroticism had a statistically significant effect with a medium effect size that emotionally stable students acquired more literal knowledge than neurotic students in an online but not traditional setting. - Emotional stability versus neuroticism had a statistically significant effect, with a large effect size, indicating more enjoyment from learning among emotionally stable students than among neurotic students. This was true in the face-to-face and two-way videoconferencing conditions. - In the one-way videoconferencing condition, neurotic personality types enjoyed learning to a greater extent than emotionally stable students.
A Blended Education Program Based on Critical Thinking and	Participants: N=41; 85% female; Age: majority 20-22 Inclusion criteria: Students of Jahrom Medical Sciences	Study Description: For both teaching formats, curriculum based on critical thinking was prepared by an integration of	-Neo Personality Test (Neo FFI)- -Student's final score	RESULTS: -Academic achievement in the blended education group was significantly higher than that of the traditional education -Type of teaching, student 's final

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Its Effect on Personality Type and Attributions Mosalanejad et al., 2010 Iran Level 4	University with the degree of B.A, registered in the course of psychological diseases in the first semester of 2008–2009. Study setting: Jahrom Medical Sciences University in Iran	different teaching methods such as face-to-face, various methods of teaching to improve students 'critical thinking (group discussion , learning based on problem solving, simulations , practical projects) and electronic learning (asynchronous). Intervention 1: (n=NR) Blended learning Intervention 2: (n=NR) Traditional learning		scores and personality type had a significant relationship between all personality factors (openness, extraversion, agreeableness, and conscientiousness). -Those who were in the neuroticism group did not correlate with type of teaching and student scores.
Sub-Theme 2: MBTI Personality Assessment				
Student Satisfaction with Blended and Online Courses Based on Personality Type Bolliger & Erichsen, 2013 United States Level: 4 cross sectional	Participants: N=72 68.1% female; age 22-64, M=38 Inclusion criteria: Graduate students enrolled in 21 blended and online courses at two research universities in the US (education and instructional technology courses) who volunteered for the study Study setting: Two	Study Description: to investigate graduate students' perceived satisfaction in blended and online learning courses based on personality type. Intervention 1: n=46 students in blended course -Site 1 -16- week semester and included required weekly 1 1/2-to-2-hour teleconference sessions. - Site 2 - varying from 8 to 16 semester weeks.	-Short, online version of the MBTI - Satisfaction questionnaire which was modified from work developed by Bolliger and Halupa (2012) -Demographics	Results: -Participants were generally satisfied with courses in both online and blended learning environments. -Majority of online students (80.8%) said they could not have taken the course in a different format, compared to 34.8% of blended learners. -Extraverts were less likely than introverts to recommend the course, -Sensors in online courses were more satisfied with working independently than intuitive learners. -Feelers and intuitives in online courses were more satisfied with threaded discussions and interactions

	research university in the US	The courses included a minimum of 8, 2.5-hour meetings via video conferencing, which were supplemented with the CMS incorporating weekly online discussion forum and communication, and outside, asynchronous collaborative activities, content management, assessment, and additional readings and instructional materials. Intervention 2: n=26 students in online course - delivered during a span of 16-weeks via the eCollege CMS. Instructors utilized the following integrated CMS application tools: information (announcements), content management (multimedia, Microsoft Office documents), assessment (electronic grade book), and communication and interaction tools (discussion boards, e-mail). Instructional materials were primarily		-introverts and intuitives expressed more dissatisfaction with interaction in blended courses.

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		text based, and students communicated and interacted asynchronously with peers and the instructor; no required class meetings were held.		
Theme 3: Evaluating the interaction of personality and choice of class type				
Impact of Personality on Choice of Instructional Delivery and Students' Performance Chesser et al, 2020 United States Level: 4 cross sectional	Participants: N=188; Mean age: 20.4 and age range: 18-51; 69% females Inclusion criteria: Undergraduate Students enrolled in "Introduction to Psychology" course from a southeastern university in the US. Study setting: Regional university in southeastern US.	Study Description: Designed to investigate how college students' personality characteristics relate to their choice of instructional delivery (online or face-to-face) and whether this choice mediates the relationship between personality and academic performance. -Undergraduate students chose to take part in either online (n=114) or in person (n=74) sections of an Introduction to Psychology course. Both courses were taught by the same professor and utilized the same resources	-Big Five Inventory (BFI; John & Srivastava, 1999); -Enrollment choice -Course grade	Results: -Conscientiousness and choice of online course was most important in terms of grade (less important: extraversion, openness, neuroticism, agreeableness) -Conscientiousness and Agreeableness were the only personality predictors related to the choice of the online environment -Conscientiousness and grade is not mediated by course environment- they were successful in both settings. -Students who chose online had a course average 3.66 points higher than those enrolled in traditional.
Online learning adoption: effects of	Participants: N=285 (out of 335) 42.8% female; 75 % between 20 -25	Study Description: To investigate the effects of two important personality traits (i.e.	- Big Five Inventory (Adapted McCrae & Costa, 1985; only looking at	Results: - Students with high neuroticism are less likely to adopt online learning. -Student with high openness to

neuroticism, openness to experience, and perceived values Watjatrakul, 2016 Thailand Level 4 cross sectional	years old, 89% undergraduate students, 43% have some experience with online courses. Inclusion criteria: Students in university Study setting: Students from a university in Thailand	openness to experience and neuroticism) on students' intentions to adopt online learning. -Survey data from students at the university were developed to investigate the influence of the personality traits of openness to experience and neuroticism to adopt online learning.	neuroticism and openness to experience) -Student perceptions of online learning's values. -Demographics	experience are more likely to to adopt online learning
Academic Performance of Virtual Students Based on Their Personality Traits, Learning Styles And Psychological Well Being: A Prediction Tabe Bordbar, 2013 Iran Level 4 cross sectional	Participants: N=200 gender= NR; Age= NR Inclusion criteria: Final year students of an online academic program at the university. Study setting: Shiraz University	Study Description: -Random selection of final year students in an online academic program to take part in survey. -The survey collected data related to personality and semester grade to provide a description of what type of students who choose to take part in an online academic program.	-5 personality factors (NEO-FFI, Kolb's list of learning styles) (McCrae & Costa, 1985) -Semester grade	Results: -Conscientiousness was better suited to online than neuroticism.

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Supplemental Table 3

Strength of Evidence Table

Strength	Criteria
Strong	Two or more Level I studies. The available evidence usually includes consistent results from well-designed, well-conducted studies. The findings as strong and they are unlikely to be strongly called into question by the results of future studies.
Moderate	At least one Level I high-quality study or multiple moderate-quality studies (i.e. Level 2A/B, 3A/B). The available evidence is sufficient to determine the effects on health outcomes, but confidence in the estimate is constrained by such factors as: • The number, size, or quality of individual studies. • Inconsistency of findings across individual studies. As more information (other research findings) becomes available, the magnitude or direction of the observed effect could change, and this change may be large enough to alter the conclusion related to the usefulness of the intervention.
Low	Small number of low-level studies, flaws in the studies, etc. The available evidence is insufficient to assess effects on health and other outcomes of relevance to occupational therapy.

Supplemental Table 4

Risk of Bias Assessment Table

Citation	Item 1	Item 2	Item 3	Item 4	Item 5
Alkis and Taskaya Temizel, 2018					
Bahcekapili and Karaman, 2020					
Baker and Moyer, 2019					
Baruth and Cohen, 2023					
Bayram, 2008					
Bhagat, 2019					
Downs, 2019					
Idrizi, 2021					
Keller and Karau, 2013					
Pinzon and Guarnizo, 2022					
Schneiderjans and Kim, 2005					
Wu and Yu, 2023					
Bishop Clark, 2007					
Blau, 2017					
Bolliger, 2013					
Chesser, 2020					
Daughenbaugh, 2002					
Harrington, 2010					
Irani, 2004					
Mosalanejad, 2010					
Tabe Bordbar, 2013					
Watjatrakul, 2016					
Yamamoto, 2023					

Mixed Methods Appraisal Tool Items 1: Are the participants representative of the target population?; 2: Are measurements appropriate regarding both the outcome and intervention (or exposure)?; 3: Are there complete outcome data?; 4. Are the confounders accounted for in the design and analysis?; 5: During the study period, is the intervention administered (or exposure occurred) as intended?

Color Coding Green: yes; Yellow: unclear; Red: no

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